1. Optical Hardware Acquisition

Raw frame capture via Pi NoIR Camera Buffer

Diffraction grating projects spectrum onto sensor

2. Frame Preprocessing

`cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)`

Isolates 2D luminance intensity matrix $\mathbf{F}(x, y)$

New Custom Modifications

3. Vertical Row Slice Extraction (NEW)

Target Center: $x_c = 400$ Aperture Width: $M = 5$

Defines strict spatial region of interest for dispersion

4. Spatial Column Averaging (SNR Boost)

$$\mathbf{I}(y) = \frac{1}{M} \sum_{x=x_c-2}^{x_c+2} \mathbf{F}(x, y)$$

Cross-averages 5 pixel columns to squash photon shot noise

Yields +124% ($\sqrt{5}$) Signal-to-Noise multiplier

Pure 1D Array $I(y)$ vector

5. Wavelet Transformation

Mathematical convolution for baseline correction

Smooths structural noise/drift from the optimized 1D signal

De-noised Spectra

6. Simulated Annealing

Stochastic global peak identification

Locks centroid emission wavelengths to sub-pixel accuracy

Emission Centroids

7. Output Visualization & Data Export

Y-axis row to nanometer (nm) calibration mapping

Live GUI tracking and CSV dataset export